

DEFENSE STANDARDIZATION PROGRAM

CASE STUDY

Permeation Testing for Low-Volatility Substances

Permeation Measurements of Nerve Agent VX Through Materials Used in Personal Protective Equipment

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Defense Standardization Program Case Study

Permeation Testing for Low-Volatility Substances



Personal protective equipment (PPE) is used to protect individuals from hazardous substances; one familiar use is to protect first responders in situations such as hazardous chemical spills. For the Department of Defense, the hazards include chemical warfare agents; assurance is needed that PPE will protect laboratory and field personnel who may come in contact with such agents. This assurance comes from standardized testing of PPE materials, which was traditionally performed via vapor testing.

DoD employees have created, patented, and implemented a new test method and test fixture for contact hazards, versus vapor hazards: Low-Volatility Agent Permeation (LVAP). In LVAP testing, comparative tests were conducted in both vapor and contact configurations with the nerve agent VX. Approximately 20-fold more contaminant was measured using contact, rather than vapor, testing. Thus, vapor testing can underestimate the amount of agent to which a worker is exposed in a contact scenario.

As new PPE articles are developed, it is important to test each material in a manner consistent with its expected use to better characterize the protection performance. The LVAP test method and fixture enable such testing for contact hazards, and this is now endorsed as a DoD Test and Evaluation standard.

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BACKGROUND

Personal protective equipment is used to protect workers from a variety of hazards, in a wide range of environments. Items such as hard hats, respirators, and full-body suits ensure that users (construction workers, lab staff, HAZMAT responders, etc.) remain safe in the presence of hazardous substances. Users and buyers of PPE look for PPE that is certified to meet a standard for protection from a given hazard. That certification is achieved by passing rigorous testing for an article of PPE for that hazard. These tests are conducted in accordance with standards[1] for a given PPE material and hazardous substance. Typically, the testing result gives a rating or specifies how long a material will resist penetration by a hazardous substance. This laboratory testing of PPE provides guidance on how long users will be protected, and therefore it is critical to selecting or acquiring PPE and creating safety procedures.

PROBLEM

For items such as respirators, gloves, and full ensembles, PPE testing is often based on detecting vapors from a liquid contaminant that have permeated or penetrated the PPE material to measure risk. A swatch of material has a precise amount of liquid contaminant applied to the materials used to make the PPE, followed by detection of vapors on the opposite side of the material. This is an appropriate test for the nerve agent sarin, which is volatile. However, vapor analysis is not the best detection method for all contaminants; for example, the nerve agent VX is a contact hazard, not a vapor hazard, due to its low volatility. Vapor-based testing could yield dangerously misleading results. Attempts to use vapor-based methods to quantify a contact hazard can result in a 20-fold under-prediction of the potential hazard. Low volatility, coupled with the fact that PPE can be in direct contact with the skin, created the need for a quantitative contact test method.[2]

Accurate permeation testing is an essential component of comprehensive safety assessment for clothing and PPE used by soldiers and civilians, such as first responders, agricultural workers, and industrial workers. It is impossible to fully evaluate the suitability and safety of any PPE (clothing or other items) without reliable data to describe how well the item excludes chemical contaminants. Permeation testing is performed according to standards such as Test Operations Procedure (TOP) 08-2-501, "Permeation Testing of Materials with Chemical Agents or Simulants (Swatch Testing)"; ASTM F739, "Standard Test Method for Permeation of Liquids and Gases through Protective Clothing Materials under Conditions of Continuous Contact"; and ISO 6529, "Protective Clothing—Protection against Chemicals—Determination of Resistance of Protective Clothing Materials to Permeation by Liquids and Gases." The apparatus shown in Figure 1 is the standard test fixture for chemical warfare agent testing.

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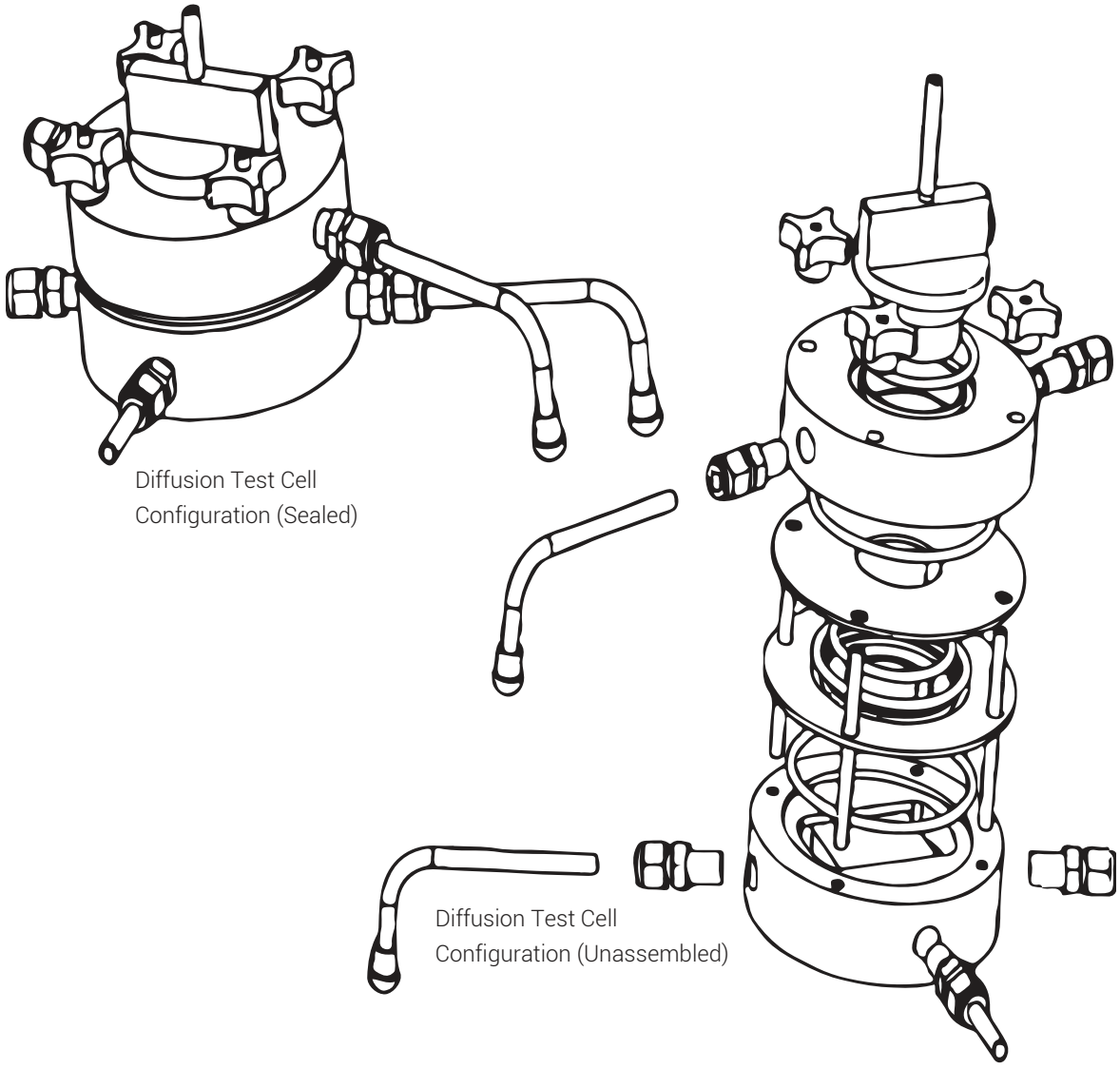
[1] Example standards are ASTM F739, "Standard Test Method for Permeation of Liquids and Gases through Protective Clothing Materials under Conditions of Continuous Contact"; BS EN ISO 6529:2001, "Protective Clothing—Protection against Chemicals—Determination of Resistance of Protective Clothing Materials to Permeation by Liquids and Gases"; NATO AEP-38, "Operational Requirements, Technical Specification, and Evaluation Criteria for NBC Protective Clothing"; and NFPA 1994, "Standard on Protective Ensembles for First Responders to Hazardous Materials Emergencies and CBRN Terrorism Incidents."

[2] All testing for verification and validation was performed with materials that are known to be permeable to VX. At no time was military PPE found deficient outside of the given specifications. No inference of vulnerability is intended or implied.

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Figure 1. Standard DoD Test Cell for Chemical Warfare Agent Permeation Testing



Source: TOP 08-2-501, "Permeation Testing of Materials with Chemical Agents or Simulants (Swatch Testing)."

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The apparatus is a two-chambered test cell. The PPE material swatch is positioned between the two halves, and the test proceeds as follows:

- A precise amount of contaminant is placed on the PPE swatch in the top section.
- The contaminant diffuses through the PPE swatch.
- A vacuum system in the bottom chamber collects the vapors that have broken through for subsequent analysis.

The use of a vapor-based test method, and the lack of a contact-based method for VX permeation, raised questions regarding laboratory safety and test readiness. To resolve this problem, the U.S. Army's Edgewood Chemical Biological Center (ECBC) designed and developed a contact-based method for permeation testing of low-volatility contaminants. The Low-Volatility Agent Permeation test fixture and test method are the result of years of research at the ECBC, supported by multiple DoD organizations.[3]

APPROACH

The response to this problem was the creation of a contact-based method to measure permeation of contaminants through PPE materials. The participants created both a new test method and a new fixture for conducting the test, and they employed standards used in laboratory testing to close a critical gap in protection research and testing.

In environmental studies, contact sorbent pads have been successfully used to quantify the uptake of contaminants from the environment; those studies benchmarked the use of these pads for organophosphorous pesticide exposures. Low-volatility organophosphorous pesticides have been used to simulate chemical warfare compounds in other studies, because nerve agents such as sarin and VX are also organophosphorus compounds.

In addition to standard fixtures, such as a preconditioning chamber, the LVAP text fixture consists of

- a test chamber, which is a commercial off-the-shelf (COTS) incubator that maintains appropriate environmental conditions during testing; and
- one or more test cells (see Figure 2), where potentially contaminated material swatches are contact tested. Each test cell is a stacked assembly of a polytetrafluoroethylene (PTFE)-lined polycarbonate Petri dish, a solid sorbent pad, a material swatch, a second PTFE layer, an O-ring gasket, and a stainless steel weight, all contained within an inverted glass jar.

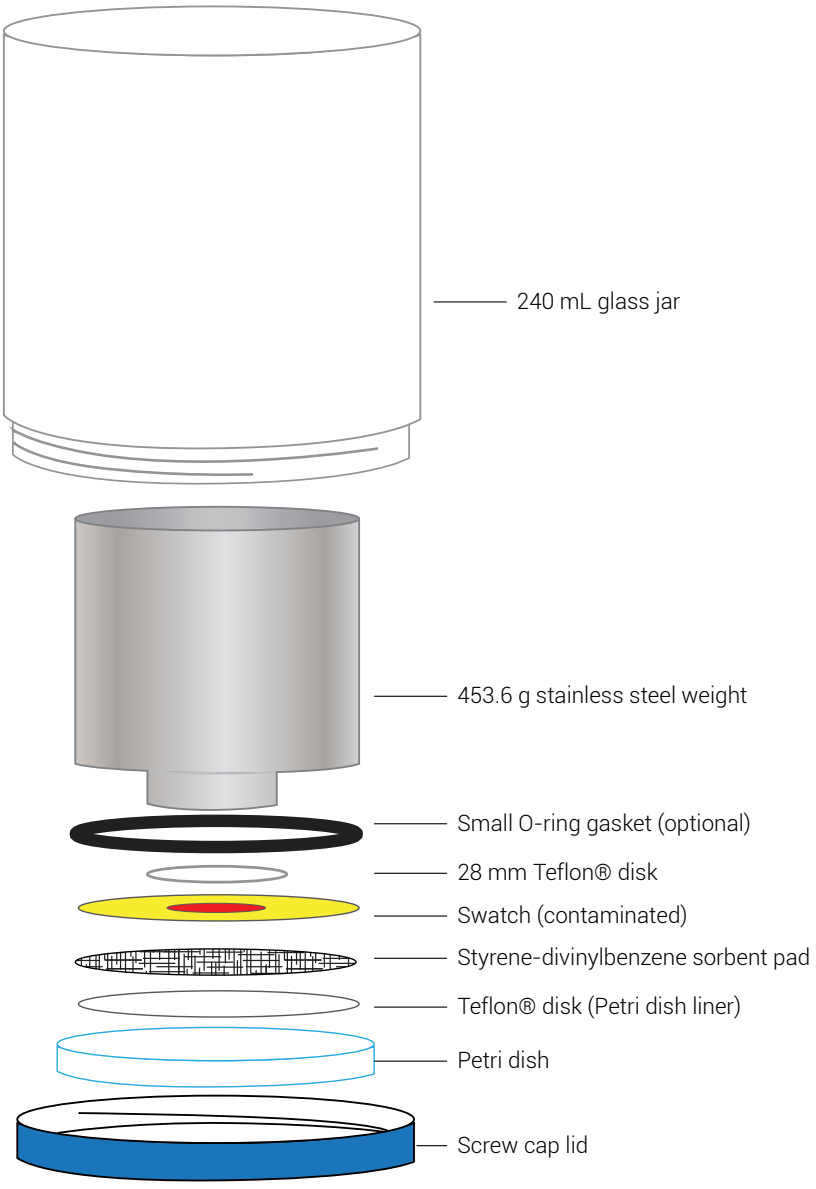
[3] Joint Science and Technology Office (Ft. Belvoir, VA), U.S. Army Natick Soldier Research, Development and Engineering Center (Natick, MA), and Joint Program Executive Office for Chemical and Biological Defense, including the Joint Project Manager for Nuclear, Biological, and Chemical Contamination Avoidance (both at Aberdeen Proving Ground, MD).

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Figure 2. Test Cell Used in LVAP Testing Fixture





U.S. government patent pending.

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The weight placed on top of the material swatch ensures that contact occurs between the swatch and sorbent pad layers. This contact is a crucial element of the system; it provides the necessary conditions for accurate measurement of agent permeation through the material. The pressure associated with the weight mass is 1 psi, which is the same pressure regime as grasping or touching objects by hand, providing additional relevancy.

For material testing, specific quantities of VX are applied to material swatches within a defined contamination area. Test cells are assembled and placed in the test chamber for a specific contact-time period. The sorbent pad from each test cell is then removed, and liquid chromatography–tandem mass spectrometry is used to quantify how much contaminant has permeated the swatch and absorbed into the pad.

During the verification and validation (V&V) testing of the new method, extensive precision and accuracy calculations were conducted and documented in accordance with International Organization for Standardization (ISO) 5725:3, "Accuracy (Trueness and Precision) of Measurement Methods and Results." These statistics provided the necessary metrics for evaluating the LVAP system as a reliable test method.

The LVAP system design is based on open architecture concepts. Given that the main LVAP components and consumables are commonly available commercial products, the LVAP system is resistant to many of the cost and technical risks associated with product obsolescence. Users can take advantage of supplier competition to minimize costs and simplify life-cycle management.



PARTICIPANTS

LVAP completed V&V under the Deputy Under Secretary of the Army for Test and Evaluation. The resulting V&V report earned O-6 level concurrence signatures from the following:

- Joint Science and Technology Office
- Joint Requirements Office for CBRN Defense
- Joint Program Executive Office for Chemical and Biological Defense
- Operational test authorities for the Army, Air Force, Navy, and Marine Corps.

The DoD Chemical and Biological Defense Program's Test and Evaluation Executive endorsed the V&V report as a test and evaluation standard, and it is now available to DoD programs of record for all services.

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At present, the LVAP system is being used at several DoD-associated facilities in support of the Uniform Integrated Protection Ensemble Family of Systems acquisition program of record. The United Kingdom recently implemented LVAP as an official test method for its own research and test program efforts.[4] The Joint CBRN Defense Capability Development Group Physical Protection Panel is updating NATO test standard AEP-38, "Operational Requirements, Technical Specification and Evaluation Criteria for NBC Protective Clothing," to include LVAP as an international test method for VX.

Partners from the Department of Homeland Security, the Environmental Protection Agency, the Federal Bureau of Investigation, and the intelligence community are considering incorporating LVAP systems into their programs to protect operators, law enforcement personnel, and first responders from low-volatility threats. ASTM International is considering LVAP as a possible new ASTM method.



STANDARDS USED

This effort benefited from many standards; almost every aspect of laboratory testing for hazardous materials is governed by one or more standards. The following is just a sample of the applicable ASTM standards:

- F739–12e1, "Standard Test Method for Permeation of Liquids and Gases through Protective Clothing Materials under Conditions of Continuous Contact"
- D471, "Test Method for Rubber Property—Effect of Liquids"
- D543, "Practices for Evaluating the Resistance of Plastics to Chemical Reagents"
- D751, "Test Methods for Coated Fabrics"
- D1777, "Test Method for Thickness of Textile Materials"
- E105, "Practice for Probability Sampling of Materials"
- E171, "Practice for Conditioning and Testing Flexible Barrier Packaging"
- E177, "Practice for Use of the Terms

[4] DSTL/TR100250, "Implementation and Validation of the Low-Volatility Agent Permeation Fixture and Method at DSTL," D'Onofrio and Pritchard, 2017.

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Precision and Bias in ASTM Test Methods"

- E691, "Practice for Conducting an Interlaboratory Study to Determine the Precision of a Test Method"
- F903, "Test Method for Resistance of Materials Used in Protective Clothing to Penetration by Liquids"
- F1001, "Guide for Selection of Chemicals to Evaluate Protective Clothing Materials"
- F1194, "Guide for Documenting the Results of Chemical Permeation Testing of Materials Used in Protective Clothing"
- F1383, "Test Method for Permeation of Liquids and Gases through Protective Clothing Materials under Conditions of Intermittent Contact"
- F1407, "Test Method for Resistance of Chemical Protective Clothing Materials to Liquid Permeation—Permeation Cup Method"
- F1494, "Terminology Relating to Protective Clothing"
- F2815, "Practice for Chemical Permeation through Protective Clothing Materials: Testing Data Analysis by Use of a Computer Program."

OUTCOME

Benefits

- First and foremost, LVAP closes a critical gap in protection research and testing for contact hazards; it enables PPE testing that is both scientifically rigorous and appropriate for contact hazards.
- The LVAP fixture costs approximately $30,000 to implement—a savings of more than 10-fold over vapor-based test systems. It actually enables low-volatility agent testing that is nearly 10-fold more precise—at one-tenth the cost.
- Its simplified, open architecture design enables a significant savings for system setup and maintenance. Key LVAP components are reusable COTS items, and required consumables are commercially sourced.
- The incubator, weight, and much of the supportive equipment are reusable. On a cost basis, approximately 93 percent of the system fixture is reusable, and all of the consumables associated with LVAP (except the chemical agent) are easily sourced COTS items.
- By focusing on contact permeation and removing the vapor collection components, the physical footprint of the fixture decreased from an 8 × 3 ft. (24 square ft.) area for vapor collection racks to a 2 × 2 ft. (4 square ft.) region for LVAP.



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- The simplified design also allows for more samples to be incorporated into the smaller region: while using the same number of operators, throughput increased from 27 samples per experiment for a vapor collection system to 40 samples per experiment for the LVAP system.
- For the same operating costs, 27 samples could be processed through the traditional method with a variability of ±80 percent and no true contact capability; alternatively, 40 samples could be processed through the LVAP method with a variability of only ±8 percent, using a proven contact-based capability that also has a smaller footprint and a lower initial setup investment.
- The LVAP system also offers advantages over standards used in earlier testing. For example, no solvents are required during the contamination period. This allows the LVAP method to be used to test air-breathable materials, and it also allows for testing within operationally relevant contact scenarios, such as when an individual grasps a contaminated object.

New and Updated Standards

As a result of this work, there are some new standards and some revisions of existing standards. The following are examples:

- The U.K. Ministry of Defence has a new test method standard—DSTL/TR100250, "Implementation and Validation of the Low-Volatility Agent Permeation Fixture and Method at DSTL."
- U.S. DoD has a new test operations procedure—TOP 08-2-503A, "Low-Volatility Agent Permeation Swatch Testing."
- NATO AEP-38 is being updated: A draft revision of "Operational Requirements, Technical Specifications and Evaluation Criteria for CBRN Protective Clothing" is under review.
- ASTM F739-12e1 will be updated.

LESSONS LEARNED

Some lessons were technical, such as the need to characterize the proportion of background signal from vapor cross-contamination, which caused a potential bias in the results of the previous designs. This problem was solved by incorporating a COTS O-ring gasket into each test cell, which effectively removed the vapor cross-contamination without interfering with permeation testing.

Some lessons were not technical; the value of collaboration and in-person communication cannot be overstated.

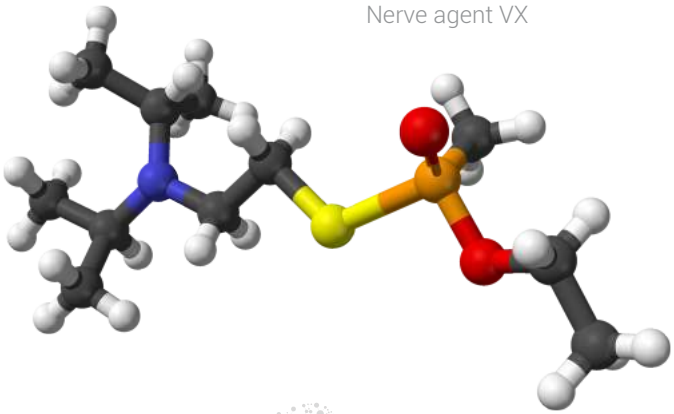
- Within the ECBC, the Operational Toxicology Branch, the Decontamination Sciences Branch, and the Permeation and Analytical Solutions Branch all worked together to provide laboratory, simulation, and leadership support.
- In DoD, the team involved the Joint Science and Technology Office, the Natick Soldier Research and Development Center, the Deputy Under Secretary of

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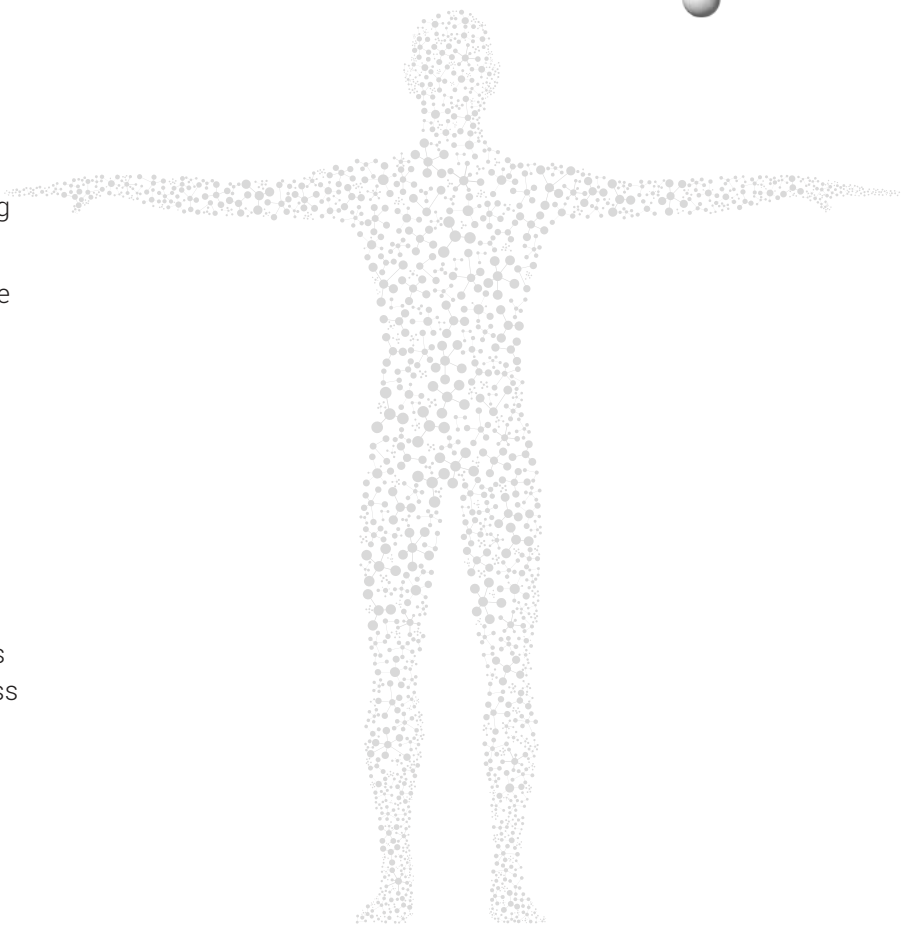


the Army for Test and Evaluation, and the Joint Project Manager for Protection.

- Beyond DoD, the team worked with the Federal Bureau of Investigation and the Department of Homeland Security.
- The team needed concurrence from a large group of diverse stakeholders within a short suspense—the accelerated timeline was required to enable LVAP to be used for upcoming programs of record. The LVAP test plan and the V&V report both required O-6 level concurrences from six organizations within DoD. The planning and testing were completed within 9 months of V&V initiation, with final concurrence and transition occurring 7 months later.
- These political challenges were overcome by open communication with all stakeholders for improved management of expectations, rigorous documentation of results, and transparency.
  - All data, including those incorporating mistakes and efforts that did not meet requirements, were clearly presented during update briefings.
  - Stakeholders and potential LVAP users were invited to the laboratory to witness testing and offer input. The personal interactions between the laboratory staff and the stakeholders helped to build a relationship of trust that ultimately enabled the success of this effort.



Nerve agent VX



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